1. Create 'spj' database'. (Database for suppliers, parts and jobs)

```
[root@localhost [(none)]> CREATE DATABASE spj;
Query OK, 1 row affected (0.002 sec)

root@localhost [(none)]>
```

2. use spj database;

```
[d1_96343_yogesh@localhost [(none)]> USE spj;
Database changed
d1_96343_yogesh@localhost [spj]>
```

3. Create following tables in spj database.

```
yogeshbhatt — mysql -u d1_96343_yogesh -p — 80×24

Records: 6  Duplicates: 0  Warnings: 0

Query OK, 0 rows affected (0.007 sec)

Query OK, 7 rows affected (0.001 sec)
Records: 7  Duplicates: 0  Warnings: 0

Query OK, 0 rows affected (0.005 sec)

Query OK, 23 rows affected (0.002 sec)
Records: 23  Duplicates: 0  Warnings: 0

[d1_96343_yogesh@localhost [spj]> SHOW TABLES;
+---------------+
| Tables_in_spj |
+---------------+
| J             |
| P             |
| S             |
| sp            |
+---------------+
4 rows in set (0.002 sec)

d1_96343_yogesh@localhost [spj]>
```

```
yogeshbhatt — mysql -u d1_96343_yogesh -p — 80×28

+-------+----------+------+-----+---------+-------+
| Field | Type     | Null | Key | Default | Extra |
+-------+----------+------+-----+---------+-------+
| J#    | char(4)  | NO   |     | NULL    |       |
| Jname | char(10) | YES  |     | NULL    |       |
| City  | char(15) | YES  |     | NULL    |       |
+-------+----------+------+-----+---------+-------+
3 rows in set (0.005 sec)

[d1_96343_yogesh@localhost [spj]> DESCRIBE P;                                ]
+--------+----------+------+-----+---------+-------+
| Field  | Type     | Null | Key | Default | Extra |
+--------+----------+------+-----+---------+-------+
| P#     | char(6)  | NO   |     | NULL    |       |
| Pname  | char(20) | YES  |     | NULL    |       |
| color  | char(6)  | YES  |     | NULL    |       |
| weight | smallint | YES  |     | NULL    |       |
| city   | char(15) | YES  |     | NULL    |       |
+--------+----------+------+-----+---------+-------+
5 rows in set (0.002 sec)

[d1_96343_yogesh@localhost [spj]> DESCRIBE S;                                ]
+--------+----------+------+-----+---------+-------+
| Field  | Type     | Null | Key | Default | Extra |
+--------+----------+------+-----+---------+-------+
| S#     | char(5)  | NO   |     | NULL    |       |
| Sname  | char(20) | YES  |     | NULL    |       |
| status | smallint | YES  |     | NULL    |       |
```

1. Create 'hr' database. Import hr-db.sql.

```
yogeshbhatt — mysql -u root -p — 80×28

yogeshbhatt@Yogesh96343D1 ~ % mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 34
Server version: 9.6.0 Homebrew

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

root@localhost [(none)]> CREATE DATABASE hr;
Query OK, 1 row affected (0.002 sec)

root@localhost [(none)]>
```

2. Create 'sales' database. Import sales-db.sql.

```
yogeshbhatt — mysql -u root -p — 80×28

root@localhost [(none)]> CREATE DATABASE sales;
Query OK, 1 row affected (0.002 sec)

root@localhost [(none)]>
```

3. List all tables in hr database. Display contents of all tables (one by one).

```
+------------+--------------------------------+------------+------------+
| AC_ACCOUNT | Public Accountant              |       4200 |       9000 |
| AC_MGR     | Accounting Manager             |       8200 |      16000 |
| AD_ASST    | Administration Assistant       |       3000 |       6000 |
| AD_PRES    | President                      |      20080 |      40000 |
| AD_VP      | Administration Vice President  |      15000 |      30000 |
| FI_ACCOUNT | Accountant                     |       4200 |       9000 |
| FI_MGR     | Finance Manager                |       8200 |      16000 |
| HR_REP     | Human Resources Representative |       4000 |       9000 |
| IT_PROG    | Programmer                     |       4000 |      10000 |
| MK_MAN     | Marketing Manager              |       9000 |      15000 |
| MK_REP     | Marketing Representative       |       4000 |       9000 |
| PR_REP     | Public Relations Representative |       4500 |      10500 |
| PU_CLERK   | Purchasing Clerk               |       2500 |       5500 |
| PU_MAN     | Purchasing Manager             |       8000 |      15000 |
| SA_MAN     | Sales Manager                  |      10000 |      20080 |
| SA_REP     | Sales Representative           |       6000 |      12008 |
| SH_CLERK   | Shipping Clerk                 |       2500 |       5500 |
| ST_CLERK   | Stock Clerk                    |       2008 |       5000 |
| ST_MAN     | Stock Manager                  |       5500 |       8500 |
+------------+--------------------------------+------------+------------+
19 rows in set (0.000 sec)

+-------------+----------------------------------------+-------------+---------------------+------------------+------------+
| LOCATION_ID | STREET_ADDRESS                         | POSTAL_CODE | CITY                | STATE_PROVINCE   | COUNTRY_ID |
+-------------+----------------------------------------+-------------+---------------------+------------------+------------+
|        1000 | 1297 Via Cola di Rie                   | 00989       | Roma                | NULL             | IT         |
|        1100 | 93091 Calle della Testa                | 10934       | Venice              | NULL             | IT         |
|        1200 | 2017 Shinjuku-ku                       | 1689        | Tokyo               | Tokyo Prefecture | JP         |
|        1300 | 9450 Kamiya-cho                        | 6823        | Hiroshima           | NULL             | JP         |
|        1400 | 2014 Jabberwocky Rd                    | 26192       | Southlake           | Texas            | US         |
|        1500 | 2011 Interiors Blvd                    | 99236       | South San Francisco | California       | US         |
|        1600 | 2007 Zagora St                         | 50090       | South Brunswick     | New Jersey       | US         |
|        1700 | 2004 Charade Rd                        | 98199       | Seattle             | Washington       | US         |
|        1800 | 147 Spadina Ave                        | M5V 2L7     | Toronto             | Ontario          | CA         |
|        1900 | 6092 Boxwood St                        | YSW 9T2     | Whitehorse          | Yukon            | CA         |
|        2000 | 40-5-12 Laogianggen                    | 190518      | Beijing             | NULL             | CN         |
|        2100 | 1298 Vileparle (E)                     | 490231      | Bombay              | Maharashtra      | IN         |
|        2200 | 12-98 Victoria Street                  | 2901        | Sydney              | New South Wales  | AU         |
|        2300 | 198 Clementi North                     | 540198      | Singapore           | NULL             | SG         |
|        2400 | 8204 Arthur St                         | NULL        | London              | NULL             | GB         |
|        2500 | Magdalen Centre, The Oxford Science Park | OX9 9ZB     | Oxford              | Oxford           | GB         |
|        2600 | 9702 Chester Road                      | 09629850293 | Stretford           | Manchester       | GB         |
|        2700 | Schwanthalerstr. 7031                  | 80925       | Munich              | Bavaria          | DE         |
|        2800 | Rua Frei Caneca 1360                   | 01307-002   | Sao Paulo           | Sao Paulo        | BR         |
|        2900 | 20 Rue des Corps-Saints                | 1730        | Geneva              | Geneve           | CH         |
|        3000 | Murtenstrasse 921                      | 3095        | Bern                | BE               | CH         |
|        3100 | Pieter Breughelstraat 837              | 3029SK      | Utrecht             | Utrecht          | NL         |
|        3200 | Mariano Escobedo 9991                  | 11932       | Mexico City         | Distrito Federal | MX         |
+-------------+----------------------------------------+-------------+---------------------+------------------+------------+
23 rows in set (0.000 sec)

+-----------+-------------+
| REGION_ID | REGION_NAME |
+-----------+-------------+
|        50 | Africa      |
|        20 | Americas    |
|        30 | Asia        |
|        10 | Europe      |
|        40 | Oceania     |
+-----------+-------------+
5 rows in set (0.000 sec)

d1_96343_yogesh@localhost [hr]>
```

4. List all tables in sales database. Display contents of all tables (one by one).

```
Query OK, 1 row affected (0.000 sec)

Query OK, 1 row affected (0.000 sec)

Query OK, 1 row affected (0.000 sec)

Query OK, 1 row affected (0.000 sec)

Query OK, 1 row affected (0.000 sec)

Query OK, 1 row affected (0.000 sec)

Query OK, 1 row affected (0.000 sec)

+-----------------+
| Tables_in_sales |
+-----------------+
| customers       |
| orders          |
| salespeople     |
+-----------------+
3 rows in set (0.001 sec)

+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2001 | Hoffman  | London   |    100 | 1001 |
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2006 | Clemens  | London   |    100 | 1001 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
| 2007 | Pereira  | Rome     |    100 | 1004 |
+------+----------+----------+--------+------+
7 rows in set (0.000 sec)

+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
10 rows in set (0.000 sec)

+------+---------+-----------+------+
| snum | sname   | city      | comm |
+------+---------+-----------+------+
| 1001 | Peel    | London    | 0.12 |
| 1002 | Serres  | San Jose  | 0.13 |
| 1004 | Motika  | London    | 0.11 |
| 1007 | Rifkin  | Barcelona | 0.15 |
| 1003 | Axelrod | New York  | 0.10 |
+------+---------+-----------+------+
5 rows in set (0.000 sec)

d1_96343_yogesh@localhost [sales]>
```

1. How many orders data is present in the database?

10 Orders

2. How many customers are present in the database?

7 Customers

3. How many salespeople are present in the database?

5 salespeople

4. On which date order with highest amount is placed?

1990-10-04

5. For which order salesman got maximum commission? Hint: you need to

refer data of two tables.
1001, Peel, comm 0.12 - onum 3011, amt 9891.88
1002, Serres, comm 0.13 - onum 3005, amt 5160.45
1004, Motika, comm 0.11 - onum 3002, amt 1900.10
1007, Rifkin, comm 0.15 - onum 3006, amt 1098.16
1003, Axelrod, comm 0.10 - onum 3009, amt 1713.23

6. Which salesmen are handling more than two customers? Hint: you need to refer data of two tables.

1001, Peel - 2001, Hoffman & 2006, Clemens
1002, Serres - 2003, Liu & 2004, Grass

7. Which city have single customer?

2004, Grass, Berlin

8. Which city has multiple salespeople?

London - 1001, Peel & 1004, Motika

9. What is name of customer and salesman of the maximum amount order?

onum 3011, Max amt 9891.88 - cnum 2006, Clemens & snum 1001, Peel

10. Study relationship between the tables in sales database

customers.snum = salespeople.snum = orders.snum;
customers.cnum = orders.cnum